



ENVOY-LAB

ISOTOPE PRODUCTION SOLUTIONS

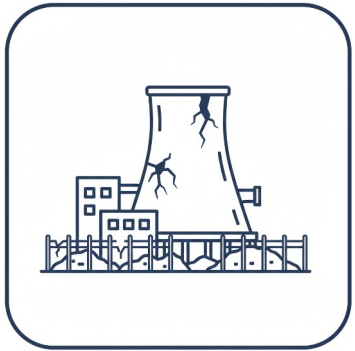
High-yield, standard cyclotron-based production system for
Actinium-225 and Lead-212

Non-Confidential Technical Summary

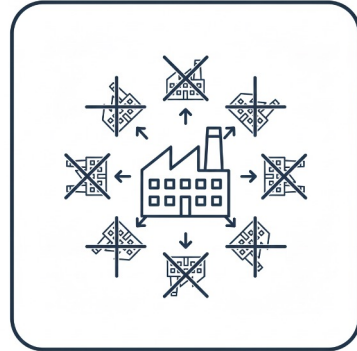
September 2026

Targeted Alpha Therapy Trials are Stalling Due to Limited Ac-225 and Pb-212

Current Production Methods



Large cyclotron or aging reactor required



Requires centralized (non-redundant) manufacturing



Inefficient production methods, difficult to scale



Ac-227 contamination co-produced in spallation



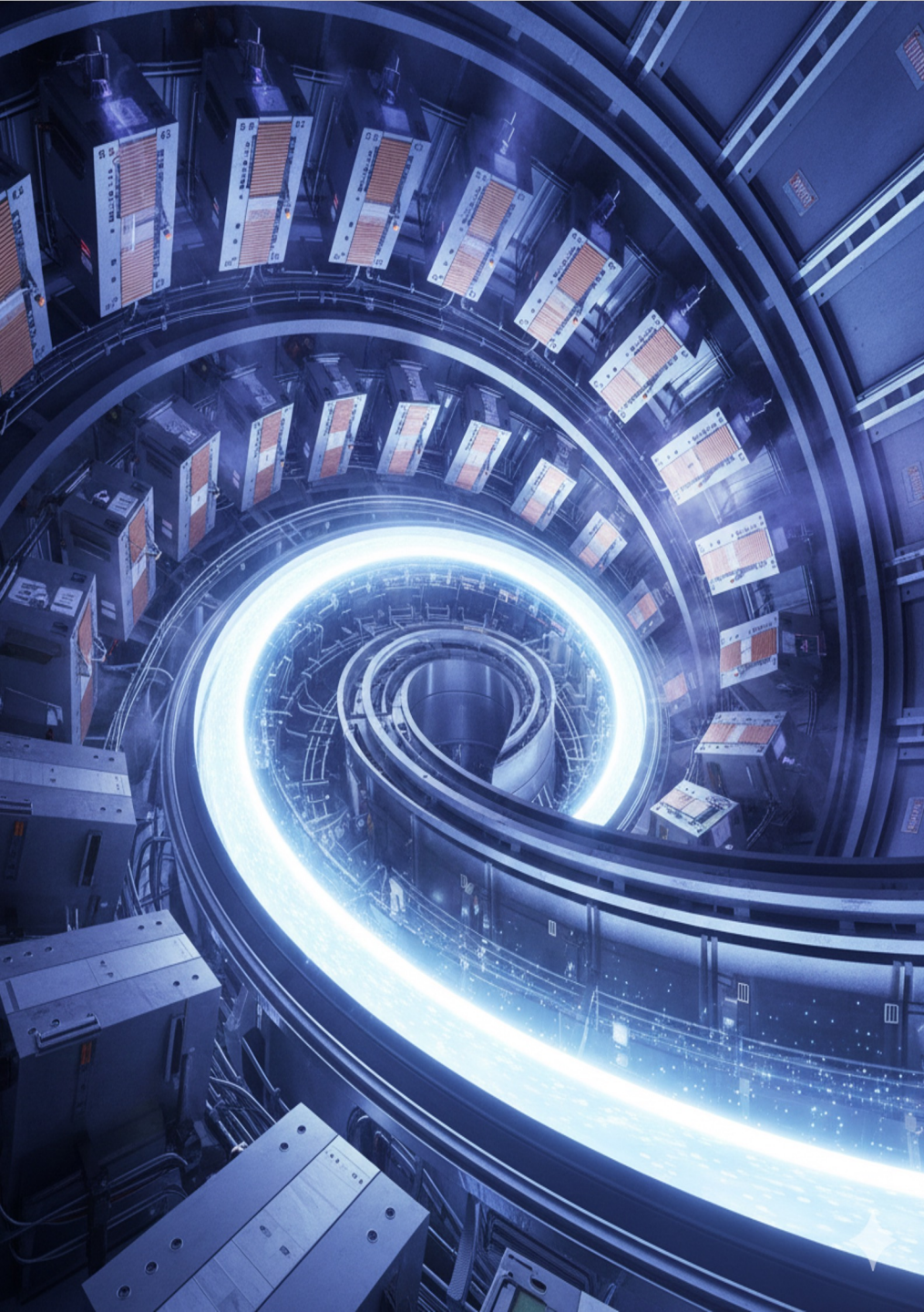
Radon release, radium fire risk in proton or gamma irradiation

EnVoy-LAB Solution

A safe, scalable, high-yield production process for
Ac-225 and Pb-212 using fast neutrons from
existing standard cyclotrons

- 18-30 MeV standard cyclotron
- Produces Ra-225 (Ac-225 precursor) & Pb-212 simultaneously or solo
 - Ra-225 yield : 18 → 120+ mCi/week
 - Pb-212 yield: 5 → 120+ mCi/day
- <1 g radium with 96-100% recovery for Ac-225
- *No radium consumption for Pb-212*
- Integrates seamlessly in existing production processes





What are we NOT?

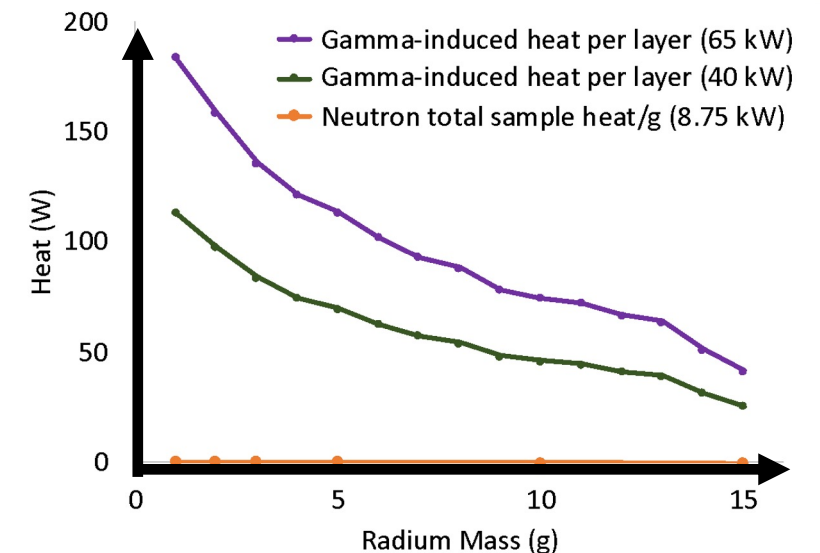
- We are NOT commissioning cyclotrons
- We are NOT building production facilities
- *We partner nonexclusively to deploy our platform to existing production facilities, increasing supply of $^{225}\text{Ac}/^{212}\text{Pb}$ as quickly and cost-effectively as possible*
- *We provide bespoke target platforms optimized for **YOUR** specific facility and radium inventory*
- *We provide technical competency to your staff in commissioning, operating, and supporting our platform*

^{225}Ac production via $^{226}\text{Ra}(n,2n)^{225}\text{Ra}$

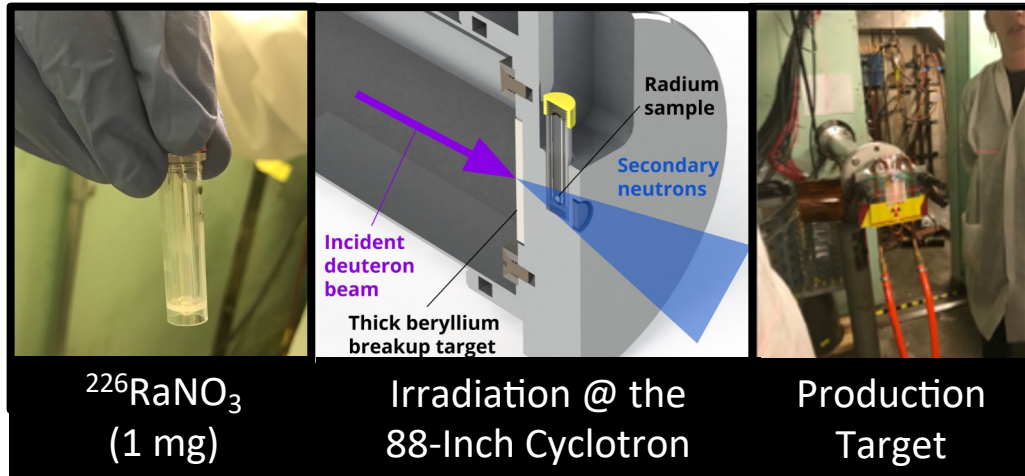
- **INDIRECTLY** irradiating ^{226}Ra with **secondary fast neutrons** from energetic protons (19-30 MeV) will induce $^{226}\text{Ra}(n,2n)^{225}\text{Ra}$
- Possible with machines as low as 16 MeV, but viable for >19 MeV
- Up to ≈ 120 mCi ^{225}Ra /week could be made with 1 g of ^{226}Ra using a high-power proton machine, using a hermetically-sealed Ra capsule
 - mCi-scale yields per week with as little as 100 mg of ^{226}Ra
- **FUNCTIONALLY NO TARGET HEATING – Safest possible operation**
 - Neutrons: ~ 8 uW/uA (of protons)
 - Photonuclear: ~ 0.1 W/uA (of electrons)
 - Protons: ~ 18 W/uA
- ^{225}Ra / ^{225}Ac separation frequency determined by client needs

We demonstrated the success of this approach in 2018 and tested an optimized breakup target in 2024

^{225}Ac 10 d	^{226}Ac 29.37 h $\beta^- = 83\%$ $\epsilon = 17\%$ $\alpha = 0.006\%$	^{227}Ac 21.77 y $\beta^- = 98.62\%$ $\alpha = 1.38\%$
^{224}Ra 3.63 d $\alpha = 100\%$ $^{14}\text{C} = 4e-9\%$	^{225}Ra 15 d	^{226}Ra 1600



^{225}Ac production: 2018 Proof-of-Concept



^{225}Ac production R&D using fast neutrons^{1,2}

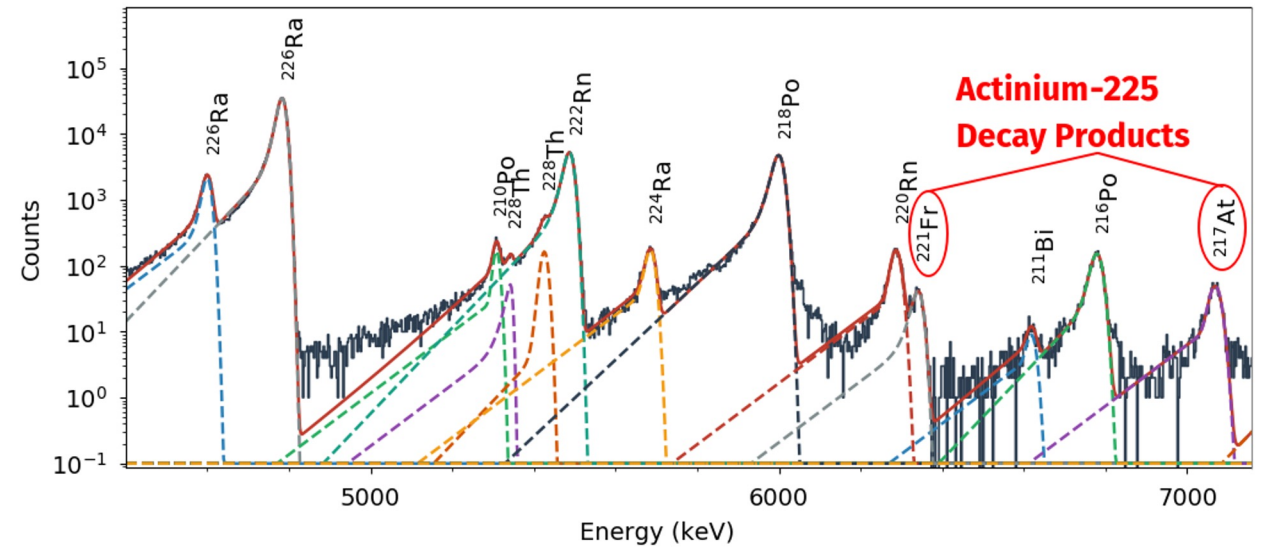
- Gamma-ray and alpha spectroscopy was performed 20 days after the end of bombardment.
- No ^{227}Ac or fission fragment contamination was observed.
- ^{225}Ac production rate at 33 MeV: $2.1 \pm 0.2 \text{ mCi/mAh/g}$
- IP owned by UC Berkeley, EnVoy has exclusive rights

¹J.T. Morrell *et al.*, [Phys. Rev. C 108, 024616 \(2023\)](#).

J. Morrell, (2021). [Next-Generation Isotope Production via Deuteron Breakup](#). :

²Lee Bernstein, Jon Batchelder, Jonathan T Morrell, Andrew Voyles. Systems and methods for producing actinium-225. WO2020210147A1. (2020)

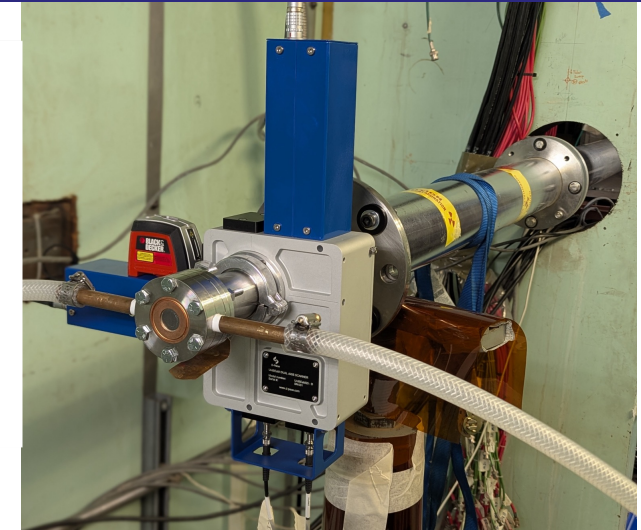
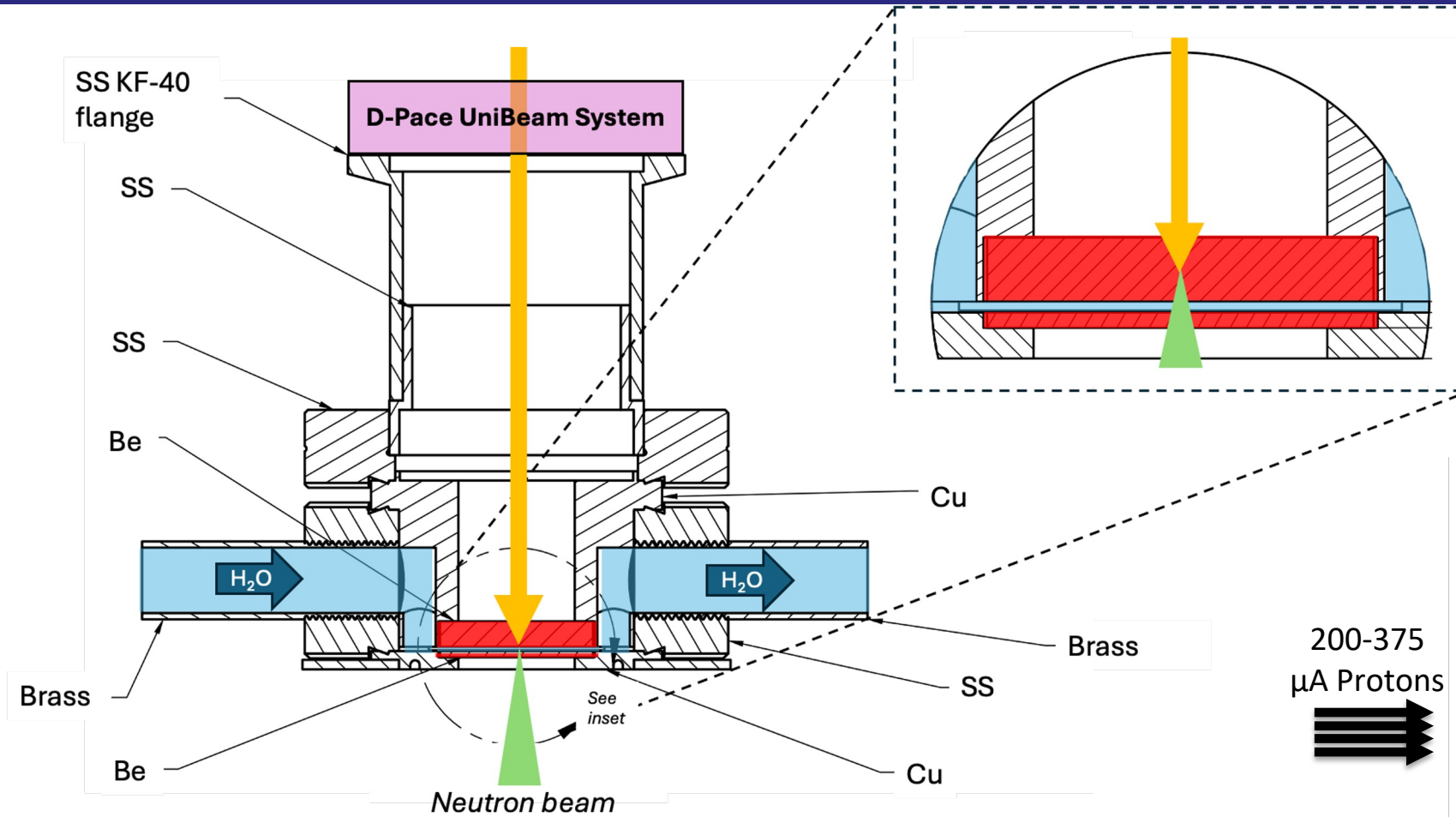
³H.J. Brede *et al.*, Nucl. Instrum. Meth. A274 (1989) 332-344



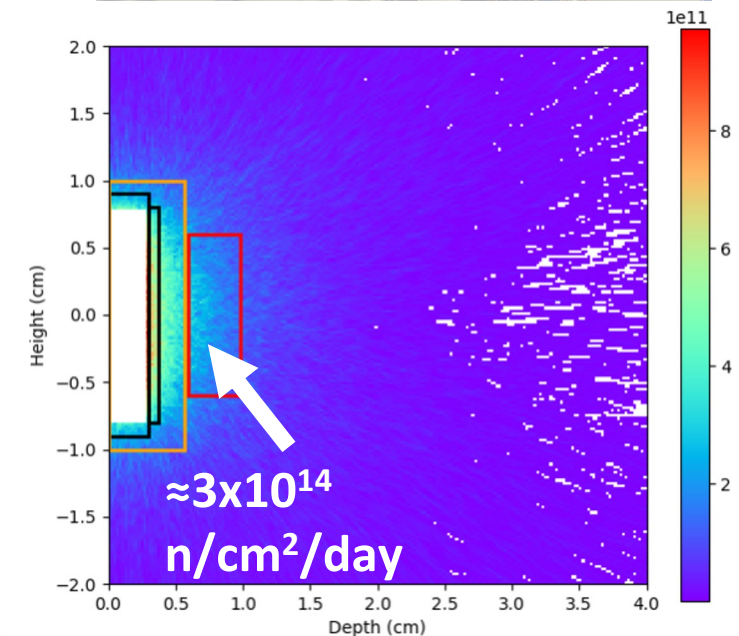
Extension to Commercialization using p+Be

- Brede *et al.*,³ has measured the neutron production from thick target *proton* irradiation of beryllium³
- We performed p+Be neutron spectral measurements using activation and and ToF spectroscopy in 2024-2025
- High-current proton machines are abundant, scalable

The EnVoy 1st Generation Target Platform



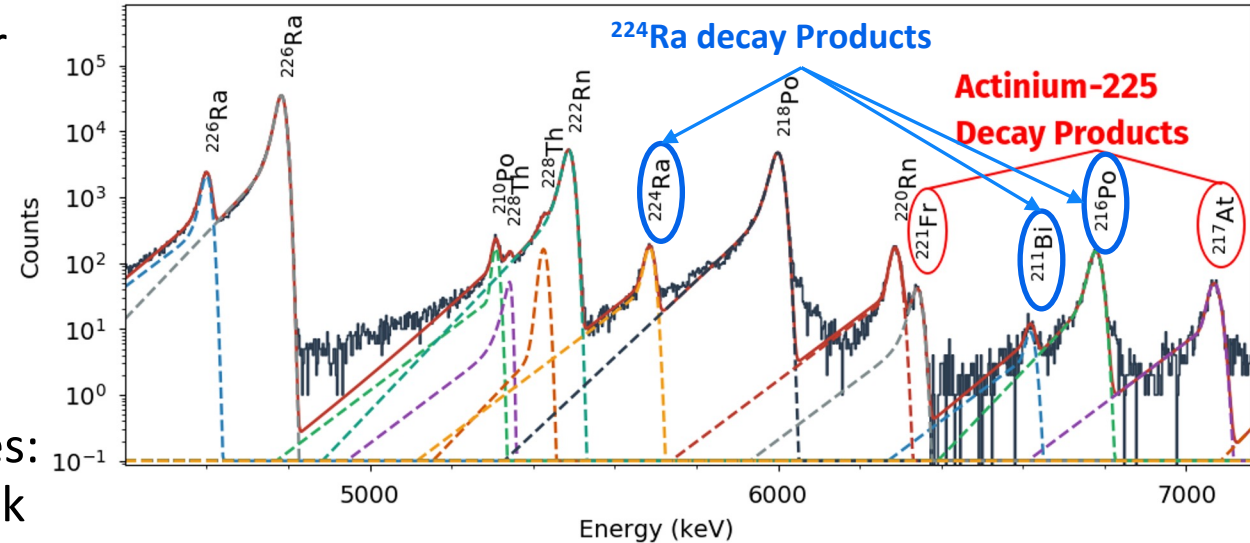
200-375
 μA Protons



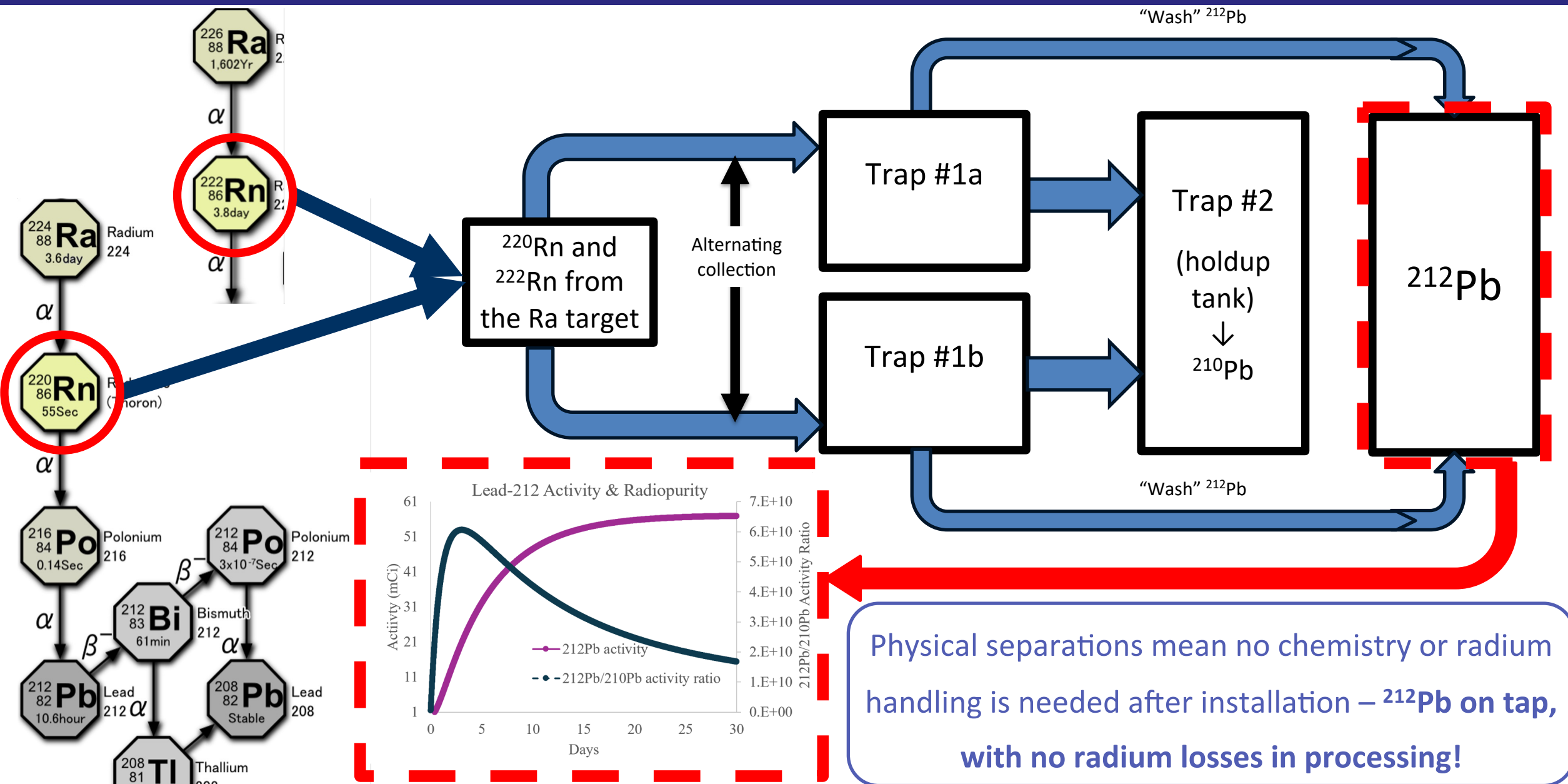
A compact, modular target platform which connects via standard vacuum flanges to commercial cyclotrons, coupled beam diagnostics

Above 19 MeV, we simultaneously make the ^{212}Pb precursor ^{224}Ra , via $^{226}\text{Ra}(n,3n)$

- The same EnVoy target platform can be operated at >19 MeV beam energies to make the ^{212}Pb precursor ^{224}Ra , via $^{226}\text{Ra}(n,3n)$
- Recoil of the ^{224}Ra ($t_{1/2} = 3.6$ d) daughter ^{220}Rn ($t_{1/2} = 55.6$ s) allows for collection via emanation, without the need for separation chemistry
- Two Ra target options for different production modes:
 - “ ^{212}Pb Only” – a sealed gas loop with holdup tank for collecting ^{212}Pb , no chemistry needed, no radium handling needed after installation
 - “Hybrid $^{212}\text{Pb}/^{225}\text{Ac}$ ” – continuous collection of ^{212}Pb during operation, and batch-mode harvest of ^{225}Ac based on user needs. Radium must be recycled prior to next irradiation cycle
- ^{212}Pb collects continuously on a glass wool cold trap – harvesting is a simple on-demand wash with dilute HCl



Pure ^{212}Pb can be harvested using a two-stage gas separation approach*



How Much ^{225}Ac , ^{212}Pb can Be Produced with the EnVoy Platform?

Beam	Max Current (uA)	Demonstrated (mCi)		Projected (mCi)		Suitable Machines
		Weekly	Daily	Weekly	Daily	
22 MeV protons @ 375 μA	385	Ra-225 Production* 59 (1g ^{226}Ra) 35 (500 mg ^{226}Ra) 9 (100 mg ^{226}Ra)	Pb-212 Production* 58 (1g ^{226}Ra) 34 (500 mg ^{226}Ra) 8 (100 mg ^{226}Ra)	Ra-225 Production* 260 (1g ^{226}Ra) 155 (500 mg ^{226}Ra) 40 (100 mg ^{226}Ra)	Pb-212 Production* 255 (1g ^{226}Ra) 150 (500 mg ^{226}Ra) 36 (100 mg ^{226}Ra)	IBA Ikon ACSI TR-Flex
19 MeV Protons @ 200 μA	446	Ra-225 Production* 18 (1g ^{226}Ra) 9 (500 mg ^{226}Ra) 3 (100 mg ^{226}Ra)	Pb-212 Production* 5 (1g ^{226}Ra) 3 (500 mg ^{226}Ra) 1 (100 mg ^{226}Ra)	Ra-225 Production* 80 (1g ^{226}Ra) 40 (500 mg ^{226}Ra) 13 (100 mg ^{226}Ra)	Pb-212 Production* 22 (1g ^{226}Ra) 13 (500 mg ^{226}Ra) 4 (100 mg ^{226}Ra)	IBA KIUBE ACSI TR-19
35 MeV Deuterons @ 50 μA	1695	Ra-225 Production* 120 (1g ^{226}Ra) 71 (500 mg ^{226}Ra) 23 (100 mg ^{226}Ra)	Pb-212 Production* 118 (1g ^{226}Ra) 69 (500 mg ^{226}Ra) 20 (100 mg ^{226}Ra)	Ra-225 Production* 1144 (1g ^{226}Ra) 682 (500 mg ^{226}Ra) 176 (100 mg ^{226}Ra)	Pb-212 Production* 1122 (1g ^{226}Ra) 660 (500 mg ^{226}Ra) 159 (100 mg ^{226}Ra)	IBA Cyclone-70 "Cyclone-XD"?

Yields at 24-30 MeV are expected to be >30+% higher than 22 MeV

Recent R&D will soon increase existing yields by a factor of ~4-15

*Illustrative values, yields may vary based on machine and local configuration

Key Platform Advantages

Compatible with widely installed machines

Use standard cyclotrons widely used in nuclear medicine facilities

Two target configurations available, both keeping radon completely sealed

Hermetically-sealed ^{225}Ac -only, or sealed-loop $^{225}\text{Ac}+^{212}\text{Pb}$

Multiple ^{225}Ac separations per irradiation plus continuous ^{212}Pb separations

The two most promising α -emitters produced *simultaneously*

High efficiency of radium, with safe, scalable solid targets

Yields scale better than linearly with radium mass, and EnVoy works with you to optimize a target for your facility

Modular target platform

User-serviceable modular target to change beam energy based on $^{225}\text{Ac}/^{212}\text{Pb}$ needs

Integrate into existing production processes and programs

An EnVoy target station can operate simultaneously to other beamlines, or independently